Engineering Pack - Merchant Dispute Resolution Portal

Run demo-20261027-081205-2ed5b3 · BRD BRD-6d8e14e4 · classification **confidential** · overall GREEN: **GREEN** (4.95/5) after 1 revision round(s) · HITL: **approved**

Quality badges

Artifact	Badge	Overall	Grounded (cited)	Coverage	Contradictions	Actionability
plan_generator	GREEN: GREEN	5.0	91%	100%	0	5.0
schedule_estimator	GREEN: GREEN	5.0	100%	100%	0	5.0
solution_architect	GREEN: GREEN	4.75	90%	100%	0	5.0
poc_planner	GREEN: GREEN	5.0	100%	100%	0	5.0
tech_stack_recommender	GREEN: GREEN	5.0	100%	100%	0	5.0

Decisions needed from the EM

• **SCH-1** (feasibility, high): p50 estimate 23.5 weeks exceeds the BRD deadline of 20 weeks (p80 28). A Must-only first release is estimated at 19 weeks. - *Default applied:* Escalated to EM: choose between extending the date, adding capacity, or a Must-only first release; schedule keeps the conservative (longer) estimate.

1. Engineering plan

Template: T1 · Team: 7 engineers

PH1 Discovery & Design (15% effort) [tmpl:T1#2]

Objective: Discovery & Design for Merchant Dispute Resolution Portal

• refined backlog with acceptance criteria

• architecture decision records

• UX flows

• integration sandbox access confirmed

• clarified acceptance criteria for ambiguous requirements NFR-06

Exit criteria: architecture review board sign-off; every Must requirement has acceptance criteria; integration credentials obtained

PH2 Foundation & Platform Setup (15% effort) [tmpl:T1#3]

Objective: Foundation & Platform Setup for Merchant Dispute Resolution Portal, delivering 1 mapped requirements *(BRD: CON-03)*

• repositories

• CI/CD pipeline

• environments (dev/stage/prod)

• IaC

• authentication skeleton

• observability baseline

• CON-03 (constraint): Must run on the org AWS platform.

Exit criteria: a hello-world service deploys to stage through the pipeline with SSO login and dashboards

PH3 Core Build (40% effort) [tmpl:T1#4]

Objective: Core Build for Merchant Dispute Resolution Portal, delivering 10 mapped requirements *(BRD: FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-09, FR-10)*

• Must-have functional requirements implemented behind feature flags with unit and contract tests

• FR-01 (must): Merchants sign in and see a list of their open and closed...

• FR-02 (must): Merchants upload evidence documents (PDF, JPG, PNG up to 20 MB each)...

• FR-03 (must): The system submits the evidence package to the card network dispute...

• FR-04 (must): Chargeback notifications from the card networks are ingested and...

• FR-05 (must): Operations analysts work a case queue, assign cases, add internal...

• FR-06 (must): Email notifications to merchants when a chargeback is created, 3...

• FR-07 (should): Analysts see a dashboard of disputes by status, reason code, win...

• FR-08 (should): Merchants can accept liability for a chargeback with one click,...

• FR-09 (must): Dispute outcomes (won or lost) are posted to the internal payments...

• FR-10 (could): Bulk evidence upload for merchants with many disputes.

Exit criteria: all Must stories accepted by product owner in stage; test coverage at least 80% on domain logic; acceptance tests pass for FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-09

PH4 Integration & Hardening (20% effort) [tmpl:T1#5]

Objective: Integration & Hardening for Merchant Dispute Resolution Portal, delivering 3 mapped requirements *(BRD: INT-01, INT-02, INT-03)*

• end-to-end integration with external systems

• performance and security testing

• NFR verification

• INT-01 (must): Visa and Mastercard dispute APIs (REST, certification required).

• INT-02 (must): Internal payments ledger API.

• INT-03 (must): Corporate email service.

Exit criteria: NFR targets met in a performance test; no open critical or high security findings; UAT sign-off

PH5 Launch & Hypercare (10% effort) [tmpl:T1#6]

Objective: Launch & Hypercare for Merchant Dispute Resolution Portal, delivering 1 mapped requirements *(BRD: CON-01)*

• production rollout plan

• runbooks

• on-call rota

• two weeks of hypercare

Exit criteria: go-live checklist complete; error budget healthy for 14 days; handover to operations

Milestones

ID	Milestone	Phase	Target (p50)	Acceptance
M1	Architecture approved [tmpl:T1#7]	PH1	end of week 3.5	architecture review board sign-off
M2	Walking skeleton in stage [tmpl:T1#7]	PH2	end of week 7	a hello-world service deploys to stage through the pipeline with SSO login and dashboards
M3	Feature complete [tmpl:T1#7]	PH3	end of week 16.5	all Must stories accepted by product owner in stage
M4	UAT sign-off and NFRs verified [tmpl:T1#7]	PH4	end of week 21	NFR targets met in a performance test
M5	Production go-live [tmpl:T1#7]	PH5	end of week 23.5	go-live checklist complete

Risks

ID	Risk	L/I	Mitigation	Owner
R1	Security review blocks launch [tmpl:T1#8]	medium/medium	threat model in Phase 1, SAST/DAST in CI, early AppSec review	Security Champion
R2	Performance NFRs missed late in the project [tmpl:T1#8]	medium/high	load-test the walking skeleton in Phase 2 and every sprint thereafter	Tech Lead

ID	Risk	L/I	Mitigation	Owner
R3	Third-party or ERP integration sandbox unavailable or late [tmpl:T1#8]	medium/medium	request sandbox in week 1, build contract-test stubs, make integration the first PoC slice	Engineering Manager
R4	INT-01 Visa and Mastercard dispute APIs (REST, certification...: sandbox access, certification or contract changes delay delivery *(BRD: INT-01)* [timeline:P-05] [tmpl:T1#2]	medium/high	Request sandbox/credentials in discovery week 1, build contract-test stubs, isolate behind an adapter (anti-corruption layer), make it a PoC slice	Tech Lead
R5	INT-02 Internal payments ledger API.: sandbox access, certification or contract changes delay delivery *(BRD: INT-02)* [tmpl:T1#2] [pbrd:PB-001#4]	medium/high	Request sandbox/credentials in discovery week 1, build contract-test stubs, isolate behind an adapter (anti-corruption layer), make it a PoC slice	Tech Lead
R6	INT-03 Corporate email service.: sandbox access, certification or contract changes delay delivery *(BRD: INT-03)* [tmpl:T1#2] [pbrd:PB-001#4]	medium/high	Request sandbox/credentials in discovery week 1, build contract-test stubs, isolate behind an adapter (anti-corruption layer), make it a PoC slice	Tech Lead
R7	Ambiguous requirement NFR-06 causes rework *(BRD: NFR-06)* [tmpl:T1#8]	medium/medium	Acceptance criteria agreed with product owner before build; conservative interpretation recorded	Product Manager

Team

- 1 x Tech Lead - owns architecture decisions and integration design [tmpl:T1#2]
- 1 x Frontend Engineer - builds the user-facing UI requirements
- 1 x QA / Test Automation Engineer - contract, end-to-end and performance test automation [tmpl:T1#5]
- 1 x DevOps / SRE Engineer - CI/CD, IaC, observability and SLO alerting [tmpl:T1#3]
- 3 x Backend Engineer - domain services and integrations

Reflection self-review: 0 issue(s) found; no changes needed

2. Schedule

p50 **23.5 weeks**, p80 **28 weeks**, 164 person-weeks, complexity basis 4, deadline 20 weeks -> DOES NOT FIT

Phase	Weeks	Start-End	Effort (pw)	Roles	Basis
PH1 Discovery & Design	3.5	w0-w3.5	24.6	Tech Lead, Product Manager	15% of 164 person-weeks; 8.9 person-weeks per requirement (median of analog projects P-05, P-06, P-11, complexity basis 4) with 7 engineers [timeline:P-05] [timeline:P-06]

Phase	Weeks	Start-End	Effort (pw)	Roles	Basis
PH2 Foundation & Platform Setup	3.5	w3.5-w7	24.6	DevOps / SRE Engineer, Tech Lead, Backend Engineer	15% of 164 person-weeks; 8.9 person-weeks per requirement (median of analog projects P-05, P-06, P-11, complexity basis 4) with 7 engineers [timeline:P-05] [timeline:P-06]
PH3 Core Build	9.5	w7-w16.5	65.6	Backend Engineer, Frontend Engineer, QA / Test Automation Engineer	40% of 164 person-weeks; 8.9 person-weeks per requirement (median of analog projects P-05, P-06, P-11, complexity basis 4) with 7 engineers [timeline:P-05] [timeline:P-06]
PH4 Integration & Hardening	4.5	w16.5-w21	32.8	QA / Test Automation Engineer, Backend Engineer, DevOps / SRE Engineer	20% of 164 person-weeks; 8.9 person-weeks per requirement (median of analog projects P-05, P-06, P-11, complexity basis 4) with 7 engineers [timeline:P-05] [timeline:P-06]
PH5 Launch & Hypercare	2.5	w21-w23.5	16.4	DevOps / SRE Engineer, Tech Lead	10% of 164 person-weeks; 8.9 person-weeks per requirement (median of analog projects P-05, P-06, P-11, complexity basis 4) with 7 engineers [timeline:P-05] [timeline:P-06]

Analogs: P-05 (cx 4, 19 wks, team 7, +19%), P-06 (cx 4, 21 wks, team 7, +17%), P-11 (cx 4, 17 wks, team 6, +21%)

3. Solution architecture

Pattern: Modular Monolith [pattern:modular-monolith] [pattern:search-centric] - Selected Modular Monolith (retrieval 0.11, 2 signals, one team, transactional workflow). Trade-offs: simplest to build, test and operate; one deploy pipeline. Weaker independent scaling and fault isolation; modules must be policed to avoid a big ball of mud. Can later be split along module boundaries (strangler).

Alternatives considered: Search-Centric (Index + Retrieval) (score 0.28; retrieval 0.14, 2 signals, no search requirements); Layered (N-tier) (score 0.08; retrieval 0.04, 0 signals)

Complexity: 4/5

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flowchart LR
  C1["Web Frontend (React SPA)"]
  C2["Application Service (modular)"]
  C3["Document Store (S3)"]
  C4["Notification Worker"]
  C5["Reporting Read Model"]
  C6["Visa and Mastercard dispute APIs Adapter"]
  C7["Internal payments ledger API Adapter"]
  C8["Corporate email service Adapter"]
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C9["Identity & Access (SSO/OAuth2 + RBAC)"]
C10["PostgreSQL (RDS)"]
C11["Redis Cache"]
C12["Observability (OpenTelemetry + Grafana)"]
C1 --> C2
C2 --> C10
C1 --> C9
C2 --> C6
C2 --> C7
C2 --> C8
C2 --> C3
C2 -->|async| C4
C10 -->|async| C5
C2 --> C11
```

ID	Component	Category	Responsibility	BRD
C1	Web Frontend (React SPA) [pattern:search-centric] [pattern:layered]	frontend (feature)	user-facing screens for FR-01, FR-02, FR-07, FR-10	FR-01, FR-02, FR-07, FR-10
C2	Application Service (modular) [pattern:modular-monolith] [pattern:layered]	service (feature)	domain modules and REST API for FR-01, FR-02, FR-03, FR-05, FR-08, FR-09, FR-10	FR-01, FR-02, FR-03, FR-05, FR-08, FR-09, FR-10
C3	Document Store (S3) [pattern:serverless] [pattern:search-centric]	storage (feature)	encrypted object storage for uploaded documents with malware scanning	FR-02, FR-10
C4	Notification Worker [pattern:event-driven] [pattern:modular-monolith]	worker (feature)	asynchronous email/SMS/push dispatch with templates and retries	FR-06, INT-03
C5	Reporting Read Model [pattern:cqrs]	analytics (feature)	read-optimised projections for dashboards and exports (CQRS-lite)	FR-07
C6	Visa and Mastercard dispute APIs Adapter	integration (feature)	anti-corruption layer for Visa and Mastercard dispute APIs: auth, mapping, retries, contract tests	FR-03, FR-04, FR-09, INT-01
C7	Internal payments ledger API Adapter [std:STD#4]	integration (feature)	anti-corruption layer for Internal payments ledger API: auth, mapping, retries, contract tests	FR-09, INT-02
C8	Corporate email service Adapter [std:STD#4]	integration (feature)	anti-corruption layer for Corporate email service: auth, mapping, retries, contract tests	INT-03
C9	Identity & Access (SSO/OAuth2 + RBAC) [std:STD#4]	auth (feature)	authentication via corporate SSO/OAuth2 and role-based authorisation	FR-01, NFR-03, NFR-04
C10	PostgreSQL (RDS) [pattern:event-driven] [std:STD#5]	database (feature)	system of record for domain state and the append-only audit log	NFR-05
C11	Redis Cache [pattern:cqrs]	cache (feature)	hot-read cache for latency-critical lookups	NFR-01
C12	Observability (OpenTelemetry + Grafana) [std:STD#6]	observability (enabler)	traces, metrics, structured logs without PII, SLO alerts	

NFR mapping

NFR	Mechanism	Components
NFR-01	Redis cache for hot reads, stateless services with horizontal autoscaling on latency, load test at 2x target load before launch	C1, C11, C2
NFR-02	Multi-AZ deployment with health checks and rolling deploys; retries with backoff and dead-letter handling for asynchronous work; SLO burn-rate alerts [std:STD#6] [pattern:event-driven]	C10, C2
NFR-03	Corporate SSO (OIDC) and OAuth2 for external users, RBAC enforced in the service layer, KMS encryption at rest, TLS 1.2+ in transit, secrets in Secrets Manager, card data tokenised via the payments vault [std:STD#4]	C10, C2, C3, C9
NFR-04	Corporate SSO (OIDC) and OAuth2 for external users, RBAC enforced in the service layer, KMS encryption at rest, TLS 1.2+ in transit, secrets in Secrets Manager, card data tokenised via the payments vault [std:STD#4]	C10, C2, C3, C9
NFR-05	Append-only audit log table with retention policy and immutable backups; access logged and reviewed [pattern:event-driven]	C10, C2
NFR-06	Org design-system components and moderated usability testing of the primary flow [std:STD#1]	C1

4. Proof of concept

Objective: De-risk Visa and Mastercard dispute APIs Adapter, Internal payments ledger API Adapter, Corporate email service Adapter, Redis Cache before Core Build by proving the riskiest requirements against measurable targets.

Duration: 3 weeks · Components: C6, C7, C8, C11

ID	Metric	Target	Measurement	Measurable
SC-NFR-01	Portal pages load in under 2 seconds at p95 for 500 concurrent... [tmpl:T2#3]	under 2 seconds at p95; 500 concurrent merchants	load test with k6 in a production-like environment	yes
SC-NFR-02	99.9% monthly availability for the merchant portal. [std:STD#5]	99.9%	synthetic probes over a 7-day soak test	yes
SC-NFR-06	primary-flow task success [tmpl:T2#9] [tmpl:T2#3]	>= 90% task completion and median time-on-task <= 3 min with 5 target users	moderated usability test with the product owner	yes
SC-INT-01	End-to-end round trip with Visa and Mastercard dispute APIs sandbox [tmpl:T2#9]	>= 95% of 50 contract-test cases pass; p95 < 1 s	contract tests against the sandbox in CI	yes
SC-INT-02	End-to-end round trip with Internal payments ledger API sandbox [tmpl:T2#9]	>= 95% of 50 contract-test cases pass; p95 < 1 s	contract tests against the sandbox in CI	yes
SC-INT-03	End-to-end round trip with Corporate email service sandbox [tmpl:T2#9]	>= 95% of 50 contract-test cases pass; p95 < 1 s	contract tests against the sandbox in CI	yes

Go / no-go: Go to Core Build if every measurable Must criterion is met; otherwise the Solution Architect revisits the pattern/stack and the EM re-baselines the schedule.

5. Tech stack options

Weights: {'scalability': 0.2, 'team_familiarity': 0.35, 'integration_risk': 0.25, 'cost': 0.2}

Option	Scal.	Familiarity	Integr. risk	Cost	Score	Approved
(rec) OPT-A Org golden path (containers + managed AWS data services)	4	5	2	3	4.15	yes
OPT-B Serverless-first (Lambda + managed services)	4	3	3	2	3.4	yes
OPT-C Python lean stack	3	4	2	3	3.6	yes

OPT-A Org golden path (containers + managed AWS data services) [std:STD#1] [std:STD#2] [decision:TD-14] - backend: Java 21 + Spring Boot 3, frontend: React 18 + TypeScript, database: PostgreSQL 16 (RDS), runtime: ECS Fargate, iac: Terraform, cache: Redis (ElastiCache), messaging: Amazon SQS/SNS, storage: Amazon S3

Pros: high team familiarity (avg 4.4/5 on the org matrix); entirely on the org approved list; no ARB exception needed; comfortable headroom for growth

Cons: standard operational overhead of containers

OPT-B Serverless-first (Lambda + managed services) [std:STD#1] [decision:TD-13] [decision:TD-02] - backend: AWS Lambda + API Gateway, frontend: React 18 + TypeScript, database: Amazon DynamoDB, runtime: AWS Lambda + API Gateway, iac: Terraform, cache: Redis (ElastiCache), messaging: Amazon SQS/SNS, storage: Amazon S3

Pros: entirely on the org approved list; no ARB exception needed; comfortable headroom for growth; pay-per-use cost at low steady volume

Cons: key-value store is a poor fit for relational audit/reporting queries (org default is PostgreSQL); low team familiarity with Amazon DynamoDB (matrix <= 2); cold starts and vendor lock-in; harder local testing

OPT-C Python lean stack [std:STD#2] [std:STD#1] - backend: Python 3.12 + FastAPI, frontend: React 18 + TypeScript, database: PostgreSQL 16 (RDS), runtime: ECS Fargate, iac: Terraform, cache: Redis (ElastiCache), messaging: Amazon SQS/SNS, storage: Amazon S3

Pros: high team familiarity (avg 4.3/5 on the org matrix); entirely on the org approved list; no ARB exception needed

Cons: least headroom if volume grows beyond the BRD

Recommendation: Org golden path (containers + managed AWS data services) has the best weighted score (4.15) with weights {'scalability': 0.2, 'team_familiarity': 0.35, 'integration_risk': 0.25, 'cost': 0.2}: scalability 4/5, familiarity 5/5, integration risk 2/5, cost 3/5.

Tools the agent chose: standards_lookup(approved technology stack team familiarity matrix scores) - approved stack and team familiarity matrix are always required; decision_log_lookup(read-model decision outcome) - architecture has a analytics component; decision_log_lookup(cache decision outcome) - architecture has a cache component; decision_log_lookup(database decision outcome) - architecture has a database component; decision_log_lookup(frontend decision outcome) - architecture has a frontend component; decision_log_lookup(integration decision outcome) - architecture has a integration component; decision_log_lookup(backend decision outcome) - architecture has a service component; decision_log_lookup(serverless decision outcome) - architecture has a worker component

Assumptions

- Phases are sequential; parallelising Foundation with late Discovery can save ~1-2 weeks.
- NFR-06 is vague (['easy', 'easy to use']); usability target deferred to PoC/product owner.
- Deployed on the org AWS platform per BRD constraint.
- NFR-06 quantified conservatively as '>= 90% task completion and median time-on-task <= 3 min with 5 target users' (org default/proxy); confirm with product owner.

Sources (retrieved knowledge-base chunks)

- decision:TD-02 - TD-02: Tech decision TD-02 (2024-03, project P-01, category database): chose PostgreSQL 16 (Amazon RDS) over MongoDB.
- decision:TD-13 - TD-13: Tech decision TD-13 (2025-02, project P-09, category serverless): chose AWS Lambda for partner webhooks over E
- decision:TD-14 - TD-14: Tech decision TD-14 (2025-10, project P-05, category frontend): chose React 18 + TypeScript with org design sy
- pattern:cqrs - Architecture Pattern Library: CQRS (Command Query Responsibility Segregation)
- pattern:event-driven - Architecture Pattern Library: Event-Driven Architecture
- pattern:layered - Architecture Pattern Library: Layered (N-tier)
- pattern:modular-monolith - Architecture Pattern Library: Modular Monolith
- pattern:search-centric - Architecture Pattern Library: Search-Centric (Index + Retrieval)
- pattern:serverless - Architecture Pattern Library: Serverless (Function-as-a-Service)
- pbrd:PB-001#4 - Vendor Invoice Approval Workflow: Vendor Invoice Approval Workflow (Past BRD) > Delivery Retrospective
- std:STD#1 - Org Engineering Standards: Org Engineering Standards (v2026.1) > Approved Technology Stack
- std:STD#2 - Org Engineering Standards: Org Engineering Standards (v2026.1) > Team Familiarity Matrix
- std:STD#4 - Org Engineering Standards: Org Engineering Standards (v2026.1) > Security and Privacy Standards
- std:STD#5 - Org Engineering Standards: Org Engineering Standards (v2026.1) > Default Service Level Objectives
- std:STD#6 - Org Engineering Standards: Org Engineering Standards (v2026.1) > Observability Standards
- timeline:P-05 - P-05: Past project P-05 Card Chargeback Automation (domain fintech, complexity 4, pattern modular-monolith): 15 requ
- timeline:P-06 - P-06: Past project P-06 Merchant Onboarding KYC (domain fintech, complexity 4, pattern api-gateway-bff): 14 requirem
- tmpl:T1#2 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Phases > Phase 1: Discovery & Design (15% of effort)
- tmpl:T1#3 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Phases > Phase 2: Foundation & Platform Setup (15% of effort)
- tmpl:T1#4 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Phases > Phase 3: Core Build (40% of effort)

• tmpl:T1#5 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Phases > Phase 4: Integration & Hardening (20% of effort)

• tmpl:T1#6 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Phases > Phase 5: Launch & Hypercare (10% of effort)

• tmpl:T1#7 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Standard Milestones

• tmpl:T1#8 - Standard Product Delivery Plan Template: Standard Product Delivery Plan Template (T1) > Risk Catalogue

• tmpl:T2#3 - Platform & Data-Intensive Delivery Plan Template: Platform & Data-Intensive Delivery Plan Template (T2) > Phases > Phase 2: Technical PoC (15% of effort)

• tmpl:T2#9 - Platform & Data-Intensive Delivery Plan Template: Platform & Data-Intensive Delivery Plan Template (T2) > Risk Catalogue